

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	SWTID-2026-2087
Project Name	POCKET SMART AI
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the POCKET SMART AI codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent formatting across HTML, CSS, JavaScript and Python files.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for frontend, backend, data, AI services and reusable components.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names such as expenseAmount, monthlyBudget, categoryName and userId are used.
4	Function / Method Names	Functions are descriptively named	Yes	Functions use names such as addExpense(), calculateBudgetStatus() and generateAllInsight().
5	Code Comments	Inline and block comments explain logic	Yes	Important business rules, API calls and non-obvious processing are documented.
6	Modular Design	Code is split into reusable functions/modules	Yes	Expense, budget, dashboard, authentication and AI logic are separated.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common validation, formatting and API utilities are reused.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation, API/database errors and user-friendly messages are included.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Expense Management Module	JavaScript / Python	Expense entry, validation and dashboard summaries	High
2	Budget Calculation Module	Python / JavaScript	Budget status, remaining amount and usage percentage	High

3	Category & Validation Utilities	JavaScript / Python	Expense categories, input validation and formatting	High
4	Authentication Module	Python / Flask	Login/session handling across protected pages	Medium
5	AI Insight Service	Python / Gemini API	Spending analysis and personalized saving suggestions	High

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Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Files and modules are logically separated and consistently organized.
Readability	5	Meaningful names, consistent indentation and clear functions improve readability.
Reusability	5	Core expense, budget, validation and AI components are designed as reusable modules.
Documentation / Comments	4	Important processing logic and integration points are documented.
Overall Score	5	Strong maintainability focus for the PocketSmart AI codebase.

Team Responsibility - Code Quality

MONISH B - Team Lead / Integration & Review

JAGADEESHWARAN S - Frontend Structure & UI Modules

Shannon Cedric Jacob - Backend/API Modules

Preethika B - Database & Validation Modules

Naresh S - AI Integration, Testing & Documentation

Template alignment: The uploaded SmartBridge template is a 5-mark document and contains the Code Layout Checklist, Reusable Components / Modules, and Overall Code Quality Assessment sections. This PocketSmart AI document keeps those sections and adapts the content to the project.